

Solution Requirements

Date	12 July 2026
Team Id	XXXXX
Project Name	Opti Crop: Smart Agricultural Production Optimization Engine
Maximum marks	4 marks

Define Functional And non – functional requirements:

- **Functional requirements (FR)** define specific behaviours, functions or features of the systems (what the system should do).
- **Non -functional requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviour (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	User logs in securely to access the system.	High
2	Authorization Levels	Farmer can access crop prediction and recommendations.	High
3	External Interfaces	Connects to soil, weather, and crop datasets.	High
4	Transaction Processing	Processes user input and generates crop predictions.	High
5	Reporting	Displays crop recommendations and prediction results.	Medium

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Generate predictions quickly.	Response within 2 seconds
2	Scalability	Support multiple users simultaneously.	Up to 500 users
3	Security & Data Privacy	Protect user and farm data.	Secure login and encrypted data
4	Reliability & Availability	Ensure system is available most of the time.	99% uptime
5	Usability & Accessibility	Easy-to-use interface for farmers.	Simple and user-friendly design
6	Other	Easy to maintain and update.	Supports future enhancements

S.No	Requirement Category	Requirement Description	Priority
6	Business Rules	Recommends crops based on soil and weather conditions.	High



Step 2: Non-Functional Requirements (NFR)